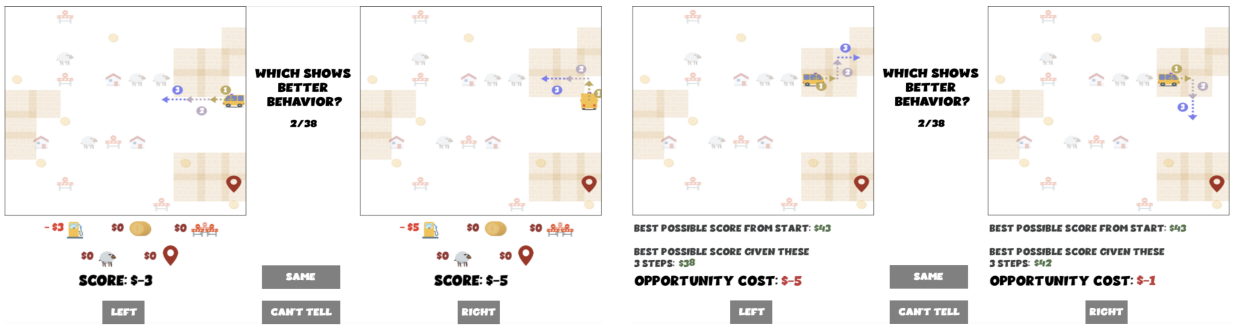


Figure 16: An example of the preference elicitation interface shown to subjects in the $P_{\Sigma_r}$-PRIVILEGED condition (Left) and the $P_{regret}$-PRIVILEGED condition (Right).

from $33/60$ subjects in the QUESTION experiment was removed: $11/20$ from the QUESTION-Control condition, $9/18$ from the $P_{\Sigma_r}$-QUESTION condition, and $13/22$ from the $P_{regret}$-QUESTION condition.

# D PRIVILEGED Experiment Interface Details

For all conditions in the PRIVILEGED experiment, when subjects interact with episodes from the grid-world domain we display four segment statistics: the "score" or partial return, the "best possible score from start" or $V_r^*(s_0^\sigma)$, the "best possible score given your moves" or $\tilde{r} + V_r^*(s_{|\sigma|}^\sigma)$, and the "opportunity cost" or regret, which is the difference between the previous two components. We explain $V_r^*(s_0^\sigma)$ as "the most money the vehicle could have made from the start", $\tilde{r} + V_r^*(s_{|\sigma|}^\sigma)$ as "the most money the vehicle can make from the start, including the route you've taken so far", and regret as "the difference between the two" and the "minimum amount of money lost by taking your route instead of the best route". See Figure 15 for an example of what humans see when interacting with an episode from the domain.

For all conditions in the PRIVILEGED experiment, we ask subjects to label preferences for $35 - 50$ segment pairs using the QUESTION "Which shows better behavior?". Only the information shown during preference elicitation differs between conditions, as detailed below.

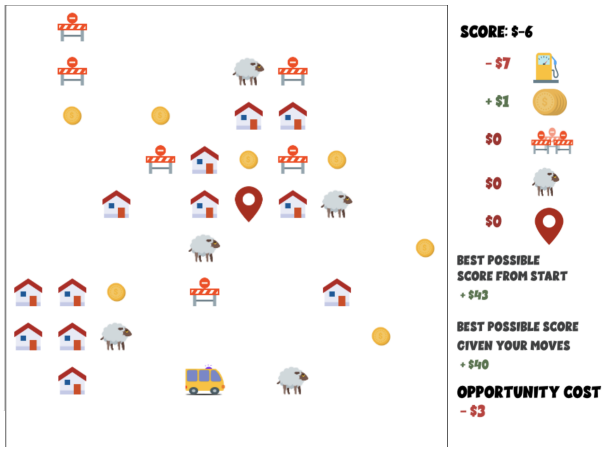


Figure 15: An episode from the grid-world domain used for teaching subjects about the domain transition and reward function, as well as to aid in their understanding of partial return (i.e., "score") and regret (i.e., "opportunity cost") in the PRIVILEGED condition.

## D.1 $P_{\Sigma_r}$-PRIVILEGED Condition Interface details

During preference elicitation, we display the "score", or the partial return, for the vehicle's path and each corresponding reward component. See Figure 16 for the preference elicitation interface for the $P_{\Sigma_r}$-PRIVILEGED condition.

## D.2 $P_{regret}$-PRIVILEGED Condition Interface details

During preference elicitation, we display the "best possible score from start" or $V_r^*(s_0^\sigma)$, the "best possible score given your moves" or $\tilde{r} + V_r^*(s_{|\sigma|}^\sigma)$, and the "opportunity cost" or regret, which is the difference between the previous two components. See Figure 16 for the preference elicitation interface for the $P_{regret}$-PRIVILEGED condition.

## D.3 PRIVILEGED-Control Condition Interface details

During preference elicitation, we do not display any segment statistics. See Figure 17 for the preference elicitation interface for the PRIVILEGED-Control condition.

Table 1: The task comprehension survey, designed to test participant's comprehension of the domain for the purpose of filtering data. Each full credit answer earned 1 point; each partial credit answer earned 0.5 points. The left most column indicates the experiment where the given QUESTION was used to filter subjects. We discarded the data of participants who scored less than 4.5 points overall for the PRIVILEGED experiment, and less than 3.5 points overall for the TRAINED and QUESTION experiments.

| Experiment | QUESTION | Full credit answer | Partial credit answer | Other answer choices |
|---|---|---|---|---|
| Privileged, Trained, QUESTION | What is the goal of this world? (Check all that apply.) | • To maximize profit | • To get to a specific location.<br>• To maximize profit<br>Partial credit was given if both answers were selected. | • To drive as far as possible to explore the world.<br>• To collect as many coins as possible.<br>• To collect as many sheep as possible.<br>• To drive sheep to a specific location. |
| Trained, QUESTION | What happens when you run into a house? | • You incur a gas cost and don't go anywhere. | • You incur a gas cost and a cost for hitting the house, and you don't go anywhere.<br>• You incur a gas cost and a cost for hitting the house, and you drive over the house.<br>• Nothing happens. | • The episode ends.<br>• You get stuck.<br>• To collect as many sheep as possible. |
| PRIVILEGED | What happens when you run into a house? (Check all that apply.) | • You pay a gas penalty.<br>• You can't run into a house; the world doesn't let you move into it.<br>Full credit was given if both answers were selected. | • You pay a gas penalty.<br>• You can't run into a house; the world doesn't let you move into it.<br>Partial credit was given if only one answer was selected. | • The episode ends.<br>• You get stuck.<br>• To collect as many sheep as possible. |
| Privileged, Trained, QUESTION | What happens when you run into a sheep? (Check all that apply.) | • The episode ends.<br>• You are penalized for running into a sheep.<br>Full credit was given if both answers were selected. | • The episode ends.<br>• You are penalized for running into a sheep.<br>Partial credit was given if only one answer was selected. | • You are rewarded for collecting a sheep. |
| Privileged, Trained, QUESTION | What happens when you run into a roadblock? (Check all that apply.) | • You pay a penalty. | | • The episode ends.<br>• You get stuck.<br>• You can't run into a roadblock; the world doesn't let you move into it. |
| Privileged, Trained, QUESTION | Is running into a roadblock ever a good choice in any town? | • Yes, in certain circumstances. | | • No. |
| PRIVILEGED | What happens when you go into the brick area? (Check all that apply.) | • You pay extra for gas. | | • The episode ends.<br>• You get stuck in the brick area.<br>• You can't go into the brick area; the world doesn't let you move into it. |
| Privileged, Trained, QUESTION | Is entering the brick area ever a good choice? | • Yes, in certain circumstances | | • No |

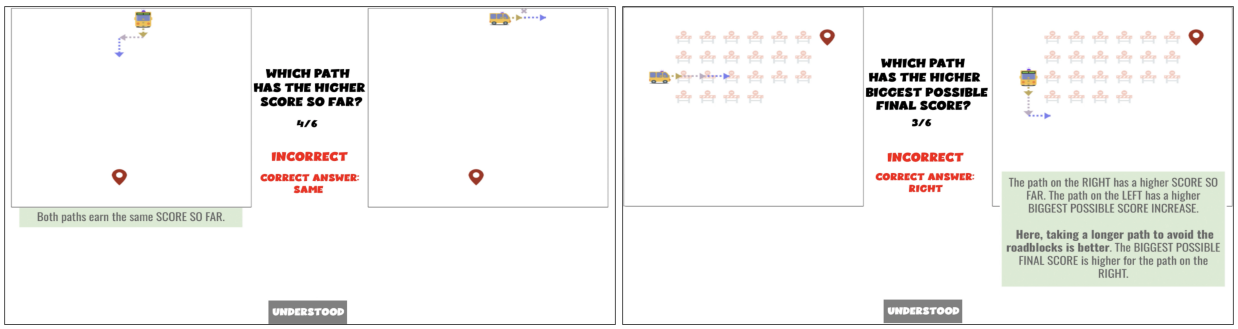


Figure 18: In the TRAINED experiment, subjects are shown two sets of six segment pairs where, after labeling their preference, they are given feedback on its correctness. For these practice segment pairs, subjects are given an explanation of why one segment is preferable to another regardless of whether their preference is correct. The left interface above displays an example of preference feedback for the partial return preference model and the right interface for the regret preference model. Preference feedback is only given for domain tasks separate from the delivery task used during the main preference elicitation session. To avoid technical jargon, we refer to partial return as "score so far", the end state value as "biggest possible score increase", and regret as "biggest possible final score". Subjects are taught these concepts during training.

# E TRAINED Experiment Interface details

The interfaces employed for each condition in the TRAINED experiment differ in what preference model—if any—human subjects' preferences are influenced towards. Therefore, the concepts taught throughout the study and the preference elicitation instruction differ between conditions, as outlined below.

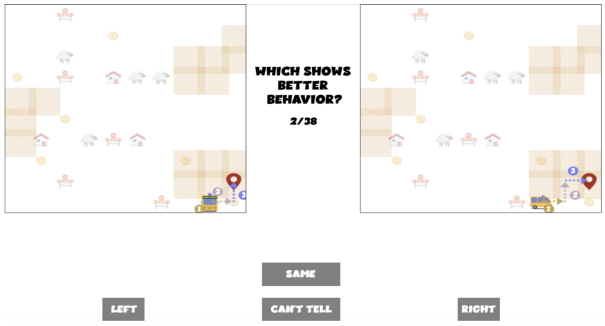


Figure 17: An example of the preference elicitation interface shown to subjects in the PRIVILEGED-Control condition.

## E.1 $P_{\Sigma_r}$-TRAINED condition Interface details

When subjects interact with episodes from the grid-world domain, we display the "score so far", or the partial return, for the vehicle's path. We explain how the "score so far" is computed, have them compute it for three trajectory segments while providing feedback, and then let them interact with additional episodes to observe the "score so far".

After understanding how to compute partial return, subjects were instructed to use it when generating preferences. Initially, they labeled preferences for six segment pairs without specific guidance. Subsequently, they were told to generate preference labels based on the "score so far" and walked through a detailed example. They then labeled six more segment pairs, receiving feedback on the correctness of their preferences. An example of this interface is shown in Figure 18. Finally, subjects were instructed on how *not* to generate preferences (e.g., "do not select the path where the van looks like it might achieve a higher score in the future"), and given another six segments pairs to label with feedback on their preferences.

The trajectory segments used to teach subjects the partial return preference model were collected from domain maps, excluding the one depicted in Figure 3, which is reserved for the main preference elicitation portion. These pedagogical segment pairs were selected to illustrate scenarios where the partial return of both segments were equal, where one segment had a higher partial return but higher regret than the other (i.e., the two competing preference models would disagree on the preference label), and where one segment had a higher partial return and lower regret than the other. The preferences collected for these segment pairs were not used for reward learning.

After learning to use the partial return preference model, subjects interacted with the map shown in Figure 3 and generated preferences for 50 segment pairs from this task. When labeling these segment pairs, subjects were asked "which path has the highest score so far?". This is the main preference elicitation phase, where no feedback or information about the ground-truth reward function is provided. The preference elicitation interface is shown in Figure 4.